Recap.

① For RVs X and Y, $E(Y|X=x) = \begin{cases} \sum_y y \cdot \frac{p_{X,Y}(x,y)}{p_X(x)} \\ \int_{-\infty}^{\infty} y \cdot \frac{p_{X,Y}(x,y)}{p_X(x)}\, dy \end{cases}$.

② For RVs X and Y, $X \perp\!\!\!\perp Y$, pdfs of X and Y are known, define $Z = X+Y$.

$$p_Z(z) = \begin{cases} \sum_y p_X(z-y)\, p_Y(y) \\ \int_{-\infty}^{\infty} p_X(z-y)\, p_Y(y)\, dy \end{cases}$$

③ For RV X with known cdf $F_X(x)$, to simulate it's distribution, define uniformly on $(0,1)$ distributed RV U, then $X = F_X^{-1}(U)$.

e.g. For $X \sim Exp(\lambda)$, $F_X(x) = 1-e^{-\lambda x}$ $(x \geq 0)$.

Let $u = F_X(x) = 1-e^{-\lambda x}$, where $U \sim Uniform(0,1)$, solve $x = -\frac{1}{\lambda}\ln(1-u)$.

Then $X = -\frac{1}{\lambda}\ln(1-U)$, since $1-U = U$, $X = -\frac{1}{\lambda}\ln U$.

Exercises.

Ex 1. Hint: $\chi_A(x,y) = \begin{cases} 1 & \text{if } (x,y)\in A \\ 0 & \text{otherwise} \end{cases}$

(i) since $e^{-y} > 0$, given pdf is non-negative.

To verify $P(\Omega)=1$: $\int_{\mathbb{R}^2} p_{X,Y}(x,y)\,dx\,dy = \int_{\mathbb{R}}\int_{\mathbb{R}} e^{-y}\chi_A(x,y)\,dx\,dy$ $\quad (0 \leq x \leq y)$

$$= \int_0^{\infty}\left(\int_x^{\infty} e^{-y}dy\right)dx$$

$$= \int_0^{\infty} e^{-x}\,dx = 1.$$

(ii) $p_Y(y) = \int_{\mathbb{R}} p_{X,Y}(x,y)\,dx = \int_0^y e^{-y}\,dx = e^{-y}\int_0^y 1\,dx = ye^{-y}$ $(y \geq 0)$

$E(X|Y=y) = \int_{\mathbb{R}} x \cdot \frac{p_{X,Y}(x,y)}{p_Y(y)}\,dx = \int_{\mathbb{R}} x \cdot \frac{e^{-y}}{ye^{-y}}\,dx = \frac{1}{y}\int_0^y x\,dx = \frac{y}{2}$, $E(X|Y) = \frac{Y}{2}$.

(iii) $p_X(x) = \int_{\mathbb{R}} p_{X,Y}(x,y)\,dy = \int_x^{\infty} e^{-y}dy = e^{-x}$ $(0 \leq x \leq y)$

$E(Y|X=x) = \int_{\mathbb{R}} y \cdot \frac{p_{X,Y}(x,y)}{p_X(x)}\,dy = \int_x^{\infty} y \cdot \frac{e^{-y}}{e^{-x}}\,dy = x+1$, $E(Y|X) = X+1$.

Ex 2. Hint: For $X \sim Exp(\lambda)$, $p_X(x) = \lambda e^{-\lambda x}$ $(x \geq 0)$.

(i) $X \perp\!\!\!\perp Y$, $p_X(x) \cdot p_Y(y) = p_{X,Y}(x,y)$. Define $A_z = \{(x,y) \in \mathbb{R}^2 \mid x+y \leq z\}$.

$$F_Z(z) = P(Z \leq z) = P(X+Y \leq z) = P((X,Y) \in A_z)$$

$$\left(= \int_{-\infty}^{z} p_Z(u)du\right) \qquad = \iint_{A_z} p_{X,Y}(x,y)\,dx\,dy$$

$$= \iint_{x+y \leq z} p_X(x) p_Y(y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{z-y} p_X(x)dx\right) p_Y(y)dy \quad (\text{define } u = x+y)$$

$$= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{z} p_X(u-y)du\right) p_Y(y)dy$$

$$= \int_{-\infty}^{z} \left(\int_{-\infty}^{\infty} p_X(u-y) p_Y(y)dy\right) du$$

Therefore $p_Z(z) = \int_{\mathbb{R}} p_X(z-y) p_Y(y) dy$.

(ii) For $X, Y \sim Exp(\lambda)$, $p_X(x) = p_Y(x) = \lambda e^{-\lambda x}$ $(x \geq 0)$

$$p_Z(z) = \int_{-\infty}^{\infty} p_X(z-y) p_Y(y) dy \qquad (z-y,\ y \geq 0)$$

$$= \int_0^z \left(\lambda e^{-\lambda(z-y)} \cdot \lambda e^{-\lambda y}\right) dy$$

$$= \lambda^2 e^{-\lambda z} \int_0^z 1\,dy = \lambda^2 z e^{-\lambda z}.$$

(iii) For $X \sim N(\mu_X, \sigma_X^2)$, $p_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$.

Define $Z = X+Y$ and use the formula from part (i).

Computing steps not discussed, see formal solution.

Ex 3. Hint: For a fixed value, consider rewrite into summation.

Geometric series $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $|x| < 1$.

(i) $$E(X) = \sum_{n=0}^{\infty} n \cdot P(X=n) \qquad \left(n = \sum_{k=1}^{n} 1\right)$$

$$= \sum_{n=0}^{\infty} \sum_{k=1}^{n} P(X=n)$$

$$= 1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + \cdots$$

$$= (P(X=1) + P(X=2) + \cdots) + (P(X=2) + P(X=3) + \cdots) + \cdots$$

$$= P(X \geq 1) + P(X \geq 2) + \cdots$$

$$= \sum_{k=1}^{\infty} P(X \geq k).$$

(ii) For $X \sim Geo(p)$, $P(X=k) = (1-p)^{k-1} \cdot p$ $(k \geq 1)$.

$$P(X \geq n) = \sum_{k=n}^{\infty} (1-p)^{k-1} \cdot p$$

$$= p \cdot (1-p)^{n-1} \sum_{k=n}^{\infty} (1-p)^{k-n} \quad (\text{define } m = k-n)$$

$$= p \cdot (1-p)^{n-1} \sum_{m=0}^{\infty} (1-p)^m \quad (|1-p| < 1)$$

$$= (1-p)^{n-1}$$

$$E(X) = \sum_{n=1}^{\infty} P(X \geq n)$$

$$= \sum_{n=1}^{\infty} (1-p)^{n-1} \quad (\text{define } m = n-1)$$

$$= \sum_{m=0}^{\infty} (1-p)^m$$

$$= \frac{1}{p}.$$

(iii) $E(X) = \int_0^\infty x\, p_X(x)\, dx \qquad \left(x = \int_0^x 1\, dy\right)$

$= \int_0^\infty \left(\int_0^x 1\, dy \right) p_X(x)\, dx \qquad (y \le x)$

$= \iint_{y \le x} p_X(x)\, dx\, dy$

$= \int_0^\infty \left(\int_y^\infty p_X(x) dx \right) dy$

$= \int_0^\infty P(X \ge y)\, dy$

$= \int_0^\infty 1 - F_X(x)\, dx$

For $X \sim Exp(\lambda)$, $F_X(x) = 1 - e^{-\lambda x}$:

$E(X) = \int_0^\infty (1 - (1 - e^{-\lambda x}))\, dx$

$= \int_0^\infty e^{-\lambda x}\, dx$

$= \frac{1}{\lambda}$.

Ex 4. Hint: Try to compute $X = F^{-1}_{X|Y=y}(U)$ for $y=0$ and $y=1$.

① $Y=0$. $\quad F_{X|Y=0}(x) = \int_0^x (2-2z)\, dz = 2x - x^2$.

Let $u = F_{X|Y=0}(x) = 2x - x^2$, solve $x = 1 \pm \sqrt{1-u}$.

Since $x \in [0,1]$, $x = 1 - \sqrt{1-u}$, $X = 1 - \sqrt{1-U}$ where $U \sim Uniform(0,1)$.

② $Y=1$, $\quad F_{X|Y=1}(x) = \int_0^x 2z\, dz = x^2$.

Let $u = F_{X|Y=1}(x) = x^2$, solve $x = \pm\sqrt{u}$.

Since $x \in [0,1]$, $x = \sqrt{u}$, $X = \sqrt{U}$ where $U \sim Uniform(0,1)$.